Data Science Concentration in Business Analytics

Carthage College

Introduction

In an increasingly data-driven business world, the ability to find meaningful insights from data was important. My three-course concentration, combined with my data science courses, helped me build a strong foundation in data science with a focus on business administration. Together, these courses gave me an understanding of how to use data science to interpret data and support valuable business decisions. I learned to identify trends and consider ways to optimize business procedures to make a meaningful impact in the business world.

Data Science Major Core Courses

The Data Science major consisted of the following 11 core courses (38 credits):

- CSC 1030: Data Science I (4 credits)
- CSC 1040: Data Science II (4 credits)
- MTH 1120: Calculus I (4 credits)
- MTH 2040: Linear Algebra (4 credits)
- MTH 3090: Mathematics of Data Science (4 credits)
- MTH 3350: Data Science Portfolio I (1 credit)
- MTH 4350: Data Science Portfolio II (1 credit)
- MTH 1050: Elementary Statistics (4 credits)
- CSC 1810: Principles of Computer Science I (4 credits)
- CSC 2810: Database Design and Management (4 credits)
- MTH 2190: Ethics in Data Science (4 credits)

My Concentration

As part of the Data Science major, students were required to complete an applied concentration to demonstrate how data science techniques could be integrated into a specific domain. This concentration helped me contextualize my technical skills in real-world applications and prepare for careers in specialized fields. I fulfilled this requirement with a concentration in Business Analytics by completing the following three courses:

MGT 3100: Introduction to Business Analytics

The course was described as a survey of mathematical models from Management Science and Operations Research, including linear programming, queuing theory, decision analysis, and simulation, applied to managerial decision-making (Carthage College 2024–2025 Catalog, p. 96).

MGT 3100 served as the foundation of my concentration by introducing key analytical tools used in business decision-making. The course exposed me to fundamental models and techniques for solving real-world problems with data. These concepts guided my understanding of how businesses could use analytics to improve efficiency, make strategic choices, and optimize operations.

MKT 3130: Marketing Principles

The course provided general knowledge of marketing, emphasizing the marketing mix (Product, Price, Promotion, and Placement) for consumer and industrial products, marketing strategies, customer behavior, and promotion. Topics included situation analysis, market segmentation, target-market identification, product and brand positioning through the 4 Ps of marketing, development and presentation of a marketing plan for a local business, and social, ethical, and legal issues in marketing (Carthage College 2024–2025 Catalog, p. 98).

MKT 3130 provided me with a foundational understanding of marketing strategies and consumer behavior, both important to business analytics. This background was valuable when interpreting data related to customer trends and considering marketing strategies.

ACC 2000: Accounting Fundamentals

The course was described as combining relevant portions of ACC 2010 Financial Accounting and ACC 2020 Managerial Accounting to provide an introduction to accounting concepts, financial statements, and the use of accounting information in decision-making. It emphasized applying financial data to planning, performance evaluation, and tactical decision-making. Topics included financial ratio analysis, capital investment evaluation, cost-volume-profit analysis, inventory valuation, and make-versus-buy analysis (Carthage College 2024–2025 Catalog, p. 24).

ACC 2000 rounded out my concentration by giving me a solid grasp of how financial data was structured, interpreted, and used in business decisions. The course helped me connect the analytical techniques learned in MGT 3100 with financial strategies and business operations. Understanding accounting data also helped me develop more accurate, data-informed models.

Conclusion

This three-course concentration in Business Analytics allowed me to apply data science principles directly to business decision-making. It strengthened my ability to analyze complex data sets and provide actionable insights for businesses. By completing the core Data Science curriculum alongside these applied business courses, I developed a strong foundation in business-focused analytical techniques.

This specialization prepared me for roles such as Business Data Analyst, Data Analyst, Business Intelligence Analyst, Operations Analyst, and Financial Analyst. In my career, I hoped to work in a role where I could help organizations make better decisions using data. I was prepared to apply these skills across sectors such as retail, finance, healthcare, and technology. I planned to use data to understand customer behavior, improve business operations, guide financial decisions, and strengthen marketing efforts. Ultimately, this concentration positioned me to contribute to the evolving field of Business Analytics and make a meaningful impact through data-driven solutions.

References

Carthage College. *Carthage College 2024–2025 Catalog*. 2025.